

ISOSAARI, Finland

WMO: 029880

Lat: 60.100N

Lon: 25.051E

Elev: 23

StdP: 14.68

Time Zone: 2.00 (EUE)

Period: 94-08

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-0.8	5.1	-5.0	4.2	-0.6	1.3	5.9	5.7	40.4	31.1	35.3	29.9	11.6	40	0.754

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.6	73.4	66.3	71.2	65.3	69.3	63.8	67.9	71.7	66.3	70.2	64.7	68.4	9.9	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.2	96.7	70.8	64.5	91.0	68.9	63.0	86.1	67.8	32.2	71.9	30.9	70.3	29.8	68.3	76.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
34.6	30.9	27.5	-3.1	76.2	5.5	3.1	-7.1	78.4	-10.2	80.2	-13.3	81.9	-17.3	84.1
			-4.6	69.6	6.1	3.0	-9.0	71.7	-12.6	73.5	-16.0	75.2	-20.4	77.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	43.0	27.8	24.0	28.6	37.5	46.4	56.6	63.2	62.7	54.4	45.3	36.7	31.7	
	DBStd	14.76	8.70	10.03	7.29	4.24	4.73	4.74	4.73	4.40	4.48	6.38	6.38	8.06	
	HDD50	3736	688	728	664	374	131	3	0	0	11	170	400	567	
	HDD65	8083	1153	1148	1129	824	577	255	91	98	317	609	850	1032	
	CDD50	1193	0	0	0	0	19	200	410	394	144	26	0	0	
	CDD65	65	0	0	0	0	0	2	36	27	0	0	0	0	
	CDH74	43	0	0	0	0	0	0	34	9	0	0	0	0	
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	
Wind	WSAvg	13.4	15.7	14.5	12.5	11.9	11.5	10.8	11.2	12.2	13.2	14.7	16.3	16.0	
Precipitation	PrecAvg	23.5	1.9	1.3	1.3	1.3	1.3	2.0	2.2	2.8	2.2	2.8	2.6	1.9	
	PrecMax	29.1	3.3	4.0	2.2	2.7	2.9	4.0	5.5	6.3	5.1	7.1	6.8	3.5	
	PrecMin	14.4	0.0	0.0	0.2	0.3	0.2	0.7	0.1	0.1	0.6	1.1	0.2	0.4	
	PrecStd	3.8	0.8	0.9	0.6	0.7	0.8	0.9	1.6	1.6	1.2	1.3	1.4	0.9	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.6	38.5	44.6	51.7	62.6	71.3	77.5	75.7	66.1	58.0	49.5	45.8	
		MCWB	40.3	36.6	39.1	44.8	53.4	63.2	68.8	68.6	60.5	55.1	47.7	44.4	
	2%	DB	39.6	36.8	40.1	47.8	58.4	68.0	74.5	72.7	63.9	56.6	48.6	44.6	
		MCWB	38.4	35.7	36.0	42.8	51.4	61.3	66.8	66.3	59.7	54.2	47.0	43.4	
	5%	DB	38.1	35.8	37.5	45.1	55.3	65.6	72.2	70.8	62.5	55.3	47.5	43.3	
		MCWB	36.9	34.8	35.0	41.3	49.0	59.2	65.8	65.1	58.8	53.5	46.0	42.0	
	10%	DB	36.6	34.9	35.8	43.2	52.9	63.5	70.2	69.1	60.9	53.9	45.5	41.1	
		MCWB	35.5	33.9	33.9	39.9	47.7	58.0	64.6	64.1	57.8	52.2	43.9	39.7	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.7	37.4	39.6	46.3	56.8	65.4	72.0	69.2	62.0	55.9	48.6	44.9
			MCDB	41.2	37.9	44.2	50.3	59.2	69.0	75.6	73.0	64.4	56.9	49.1	45.2
2%		WB	38.7	35.9	36.8	43.7	52.2	62.6	68.4	67.7	60.9	54.8	47.4	43.7	
		MCDB	39.3	36.4	38.9	46.6	56.1	66.6	72.2	71.6	63.2	56.0	48.3	44.4	
5%		WB	37.2	35.0	35.4	42.0	49.9	60.4	66.9	66.2	59.5	53.7	46.1	42.0	
		MCDB	37.9	35.6	36.9	44.6	54.2	64.1	70.9	69.9	61.8	55.2	47.5	43.0	
10%		WB	35.6	34.1	34.3	40.4	48.1	58.7	65.3	64.8	58.0	52.3	44.1	39.9	
		MCDB	36.3	34.8	35.4	42.8	52.1	62.3	69.3	68.2	60.5	54.0	45.2	41.0	
Mean Daily Temperature Range	5% DB	MDBR	6.2	7.0	7.6	6.2	7.2	6.9	6.6	6.1	5.7	4.7	5.0	5.9	
		MCDBR	5.1	4.4	7.2	9.3	10.6	9.6	8.5	7.2	6.7	3.9	4.9	3.6	
		MCWBR	5.6	4.8	4.7	6.0	6.3	5.8	5.5	4.6	5.1	3.8	5.6	4.2	
	5% WB	MCDBR	5.2	4.4	6.2	8.8	9.1	8.3	6.7	6.0	6.0	3.7	4.4	3.3	
		MCWBR	5.8	4.8	4.7	6.2	6.4	6.0	5.0	4.5	4.9	3.8	5.4	4.1	
Clear-Sky Solar Irradiance	taub	0.280	0.312	0.334	0.361	0.359	0.357	0.379	0.373	0.346	0.334	0.308	0.265		
	taud	2.146	2.194	2.246	2.332	2.410	2.480	2.441	2.456	2.520	2.396	2.218	2.096		
	Ebn at Noon	151	213	250	266	275	278	268	260	249	208	139	107		
	Edh at Noon	14	23	30	33	33	31	32	29	23	19	13	10		
All-Sky Solar Radiation	RadAvg	79	259	732	1222	1674	1740	1677	1301	792	345	96	47		
	RadStd	14	49	94	128	180	135	151	133	103	53	17	7		

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (60 neighbors)		+0.76	N/A	N/A	N/A	N/A	N/A	-252	N/A	N/A	N/A

Nomenclature: See separate page